

QUINT-DIODE/12-24DC/2X20/1X40 - Redundancy module



2320157

24VDC POWER SUPPLY REDUNDANCY MODULE

<https://www.phoenixcontact.com/us/products/2320157>

TAGS: PWS1232

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DIN rail diode module 12-24 V DC/2x20 A or 1x40 A. Uniform redundancy up to the consumer.



Product description

A safe redundant system is the result of the parallel connection of two power supply units which are decoupled from one another. To further increase system availability, QUINT DIODE provides the solution: decoupling with diode.

Your advantages

- Flexible
- Rugged design
- Consistent redundancy up to the load

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Technical data

Input data

DC operation

Nominal input voltage range	12 V DC ... 24 V DC
Input voltage range	10 V DC ... 30 V DC
Voltage type of supply voltage	DC
Reverse polarity protection	yes, < 60 V
Nominal input current (I_N)	2x 20 A (-25 °C ... 60 °C)
	1x 40 A (-25 °C ... 60 °C)
Maximum current I_{max}	2x 30 A (-25 °C ... 40 °C)
	1x 60 A (-25 °C ... 40 °C)
Transient surge protection	Varistor
Voltage drop, input/output	typ. 0.5 V
Nominal input voltage range	12 V DC ... 24 V DC
Input voltage range	10 V DC ... 30 V DC
Input voltage range DC	10 V DC ... 30 V DC

Output data

Efficiency	> 97 %
Nominal output voltage	$U_{in} - 0,5 V$
Nominal output current (I_N)	40 A (Increasing power)
	20 A (Redundancy)
Derating	60 °C ... 70 °C (2.5 %/K)
Power loss nominal load max.	10 W ($I_{OUT} = 20 A$)
Connection in series	no
Derating	60 °C ... 70 °C 2.5 %/K

Connection data

Input

Connection method	Screw connection
Conductor cross-section, rigid min.	0.2 mm ²
Conductor cross-section, rigid max.	6 mm ²
Conductor cross-section flexible min.	0.2 mm ²
Conductor cross-section flexible max.	4 mm ²
Conductor cross-section AWG min.	12
Conductor cross-section AWG max.	10
Stripping length	8 mm
Screw thread	M3
Tightening torque, min	0.5 Nm
Tightening torque max	0.6 Nm

Output

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Connection method	Screw connection
Conductor cross-section, rigid min.	0.5 mm ²
Conductor cross-section, rigid max.	16 mm ²
Conductor cross-section flexible min.	0.5 mm ²
Conductor cross-section flexible max.	16 mm ²
Conductor cross-section AWG min.	10
Conductor cross-section AWG max.	6
Stripping length	10 mm
Screw thread	M4
Tightening torque, min	1.2 Nm
Tightening torque max	1.5 Nm

Electrical properties

Insulation voltage input, output / housing	1000 V
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Product properties

Product family	QUINT DIODE
MTBF (IEC 61709, SN 29500)	40000000 h
LED	no

Insulation characteristics

Protection class	III
Degree of pollution	2

Dimensions

Width	50 mm
Height	130 mm
Depth	125 mm
Horizontal pitch	2.8 Div.

Installation dimensions

Installation distance right/left	5 mm / 5 mm
Installation distance top/bottom	50 mm / 50 mm

Mounting

Mounting type	DIN rail mounting
Assembly note	alignable: P _N ≥50%, 5 mm horizontally, 15 mm next to active components, 50 mm vertically alignable: P _N <50%, 0 mm horizontally, 40 mm vertically top, 20 mm vertically bottom
Mounting position	horizontal DIN rail NS 35, EN 60715

Material specifications

Flammability rating according to UL 94 (housing / terminal blocks)	V0
Housing material	Metal
Housing material	Steel sheet, zinc-plated

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Environmental and real-life conditions

Ambient conditions

Degree of protection	IP20
Ambient temperature (operation)	-40 °C ... 70 °C (> 60 °C Derating: 2,5 %/K)
Ambient temperature (storage/transport)	-40 °C ... 85 °C
Maximum altitude	≤ 5000 m
Climatic class	3K3 (in acc. with EN 60721)
Max. permissible relative humidity (operation)	≤ 95 % (at 25 °C, non-condensing)
Shock	18 ms, 30g, in each space direction (according to IEC 60068-2-27)
Vibration (operation)	< 15 Hz, amplitude ±2.5 mm (according to IEC 60068-2-6) 15 Hz ... 150 Hz, 2.3g, 90 min.
Temp code	T4 (-25 ... +70 °C; > 60 °C, Derating: 2,5 %/K)

Standards and regulations

Standard – Electronic equipment for use in electrical power installations and their assembly into electrical power installations	EN 50178/VDE 0160 (PELV)
Standard - Electrical safety	EN 60950-1/VDE 0805 (SELV)
Standard – Protection against shock currents, basic requirements for protective separation in electrical equipment	EN 50178
Standard – Safety extra-low voltage	IEC 60950-1 (SELV) and EN 60204-1 (PELV)
Standard - Safe isolation	DIN VDE 0100-410

Approvals

UL approvals	UL/C-UL listed UL 508
	UL/C-UL Recognized UL 60950-1
	UL ANSI/ISA-12.12.01 Class I, Division 2, Groups A, B, C, D T3C ... T4 (Hazardous Location)

Conformity/Approvals

ATEX	Ex II 3 G Ex ec IIC T4 Gc
	DEKRA 20ATEX0041 X
IECEX	Ex ec IIC T4 Gc
	IECEX DEK 20.0022X

EMC data

Electromagnetic compatibility	Conformance with EMC Directive 2014/30/EU
Low Voltage Directive	Conformance with Low Voltage Directive 2014/35/EC
EMC requirements for noise emission	EN 61000-6-3
	EN 61000-6-4
EMC requirements for noise immunity	EN 61000-6-1
	EN 61000-6-2

Surge voltage load (surge)

Standards/regulations	EN 61000-4-5
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Surge voltage load (surge)

Input	2 kV (level 3 - asymmetrical: conductor to ground)
	1 kV (Level 2 - symmetrical: Conductor to conductor)
Output	2 kV (level 3 - asymmetrical: conductor to ground)
	1 kV (Level 2 - symmetrical: Conductor to conductor)
Comments	Criterion A

Surge current load (surge)

Standards/regulations	EN 61000-4-5
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Surge current load (surge)

Input	2 kV (level 3 - asymmetrical: conductor to ground)
	1 kV (Level 2 - symmetrical: Conductor to conductor)
Output	2 kV (level 3 - asymmetrical: conductor to ground)
	1 kV (Level 2 - symmetrical: Conductor to conductor)

Emitted interference

Standards/regulations	EN 61000-6-3
Radio interference voltage in acc. with EN 55011	EN 55011 (EN 55022) Class B, area of application: Industry and residential
Emitted radio interference in acc. with EN 55011	EN 55011 (EN 55022) Class B, area of application: Industry and residential

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Phoenix Contact USA
586 Fulling Mill Road
Middletown, PA 17057, United States
(+717) 944-1300
info@phoenixcon.com